



SCHOOL OF INFORMATICS & IT

Diploma in Big Data & Analytics

**Data Mining and Business Analytics
(CDA2C02)**

**DMBA Assignment Report 1
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Declaration of Originality

I am the originator of this work and I have appropriately acknowledged all other original sources used as my references for this work.

I understand that Plagiarism is the act of taking and using the whole or any part of another person's work, including work generated by AI, and presenting it as my own.

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1. Translation of Business Problem into Data Mining Objectives

Business Problem

The HR Division are worried about an issue that is growing, employee attrition. They see staff leaving their company regardless of their performance. They can be performing very high or low, but that does not stop them from resigning. Since this seems unpredictable, the company requires more data-driven insights on the factors that can contribute towards this issue, and to take measure to identify them early and prevent this.

Data Mining Tasks

Predictive Modelling is needed as the target label is attrition, which will tell us yes or no if the employee is at risk of leaving the company. The model will help us to identify the factors with the highest impact for attrition and to ensure the HR can use it to flag high risk cases. Classification such as Logistic Regression and Decision Tree can be used as a modelling task.

Pattern Discovery is also needed as HR wants to find out about the hidden patterns behind attrition. It can help us find for any clusters for employees that may leave for similar reasons, or for any possible combinations that are more common as reasons for leaving.

Both Prediction and Pattern Discovery are required as the business problems requires information on both who and why are they leaving. Prediction will give us early warnings and the level of risks each employee is at. Pattern Discovery will give explanatory patterns that support HR decision making and valuable insights to help with targeted interventions such as training.

Unsupervised techniques such as clustering can help to segment employments into different groups such as young and stressed employees etc. Association rules are useful to identify any frequent combinations like low job satisfaction and high time. These will help to reveal patterns on top of the predictive models and provide more in-depth insights.

Data Analytics Requirement

The very first step is data preparation. Raw data needs to be cleaned. This can include handling missing values, outliers or inconsistencies. Ensure all fields are in the correct form for further analysis. After cleaning the data, we need to do feature selection to focus only on the relevant factors that are significant in predicting the binary outcome. Next, we need to handle the values as there are both numerical and categorical values. Ensure they are converted into numeric format so that it is appropriate for models. Lastly, we need to predict who is at high risk of leaving so that they can be flagged early. The reasons and hidden patterns need to be explained to guide meaningful HR actions.

2. Data Exploration and Cleaning

Data Profiling

The dataset contains 5000 employee records with 27 columns. The target variable is Attrition which contains Yes and No.

Column	Data Type	Range/Values	Any Observations?
Employee ID	String	EMP1000 – EMP2785... (5000)	
Gender	String	Male, Female	
Age	Numeric	24,55,26... (39)	
Marital Status	String	Married, Single, Divorced...	
Children	Numeric	0,1,2,3	
Education Level	String	Degree, Diploma, Master...	
Department	String	Finance, Operations, Customer Service...	Inconsistencies with IT and HR
Job Role	String	HR Executive, Sales Executive...	
Years at Company	Numeric	0,1,2... (41)	
Years in current Department	Numeric	0,1,2... (41)	
Years with Supervisor	Numeric	01,2... (39)	
Current Grade	Numeric	1,2,3... (7)	
Years at Current Grade	Numeric	0,1,2... (11)	
Monthly Income	Numeric	4236,5183... (4374)	Missing Values
Overtime Hours Per Week	Numeric	3,10,4... (27)	
Work Life Balance	Numeric	1,2,3,4,5	
Job Satisfaction	Numeric	1,2,3,4,5	
Commute Time	Numeric	62,56,27... (76)	
Training Hours Per Year	Numeric	4,23,24... (41)	

Performance Rating	Numeric	1,2,3,4,5	
Promotion Last 5 Years	Numeric	0,1,2,3,4	
Attrition	String	Yes, no	
Employee Engagement Score	Float	34.14,48,27,72.79... (3561)	
Internal Newsletter Clicks	Numeric	21,27,19... (51)	
Profile Completeness	Numeric	62,65,89... (41)	
Mentorship Program Opt in	Numeric	Yes, no	
HR App Usage Frequency	Numeric	68,14,6... (101)	

Data Quality Issues

Results - Field Summary (3) - Out - Output

22 of 22 Fields | Cell Viewer | 27 records displayed | Search | Data | Metadata | Actions | 000

Record	Name	Field Category	Min	Max	Median	Std. Dev.	Percent Missing	Unique Values	Mean	Layout
7	YearsAtCompany	Numeric	0	40	8	9.215737	0	41	10.6326	Layout
8	OvertimeHoursPerWeek	Numeric	0	30	8	5.156857	0	27	7.7344	Layout
9	MonthlyIncome	Numeric	1500	21997	10097.5	5299.838169	0.24	4374	10440.537089	Layout
10	YearsWithSupervisor	Numeric	0	39	3	6.305752	0	39	5.343	Layout
11	ProfileCompleteness	Numeric	60	100	79	11.940326	0	41	79.644	Layout
12	HRAppUsageFrequency	Numeric	0	100	50	29.31254	0	101	49.8516	Layout
13	YearsInCurrentDept	Numeric	0	40	3	6.373002	0	41	5.2778	Layout
14	InternalNewsletterClicks	Numeric	0	50	25	14.583277	0	51	25.2996	Layout

← Department

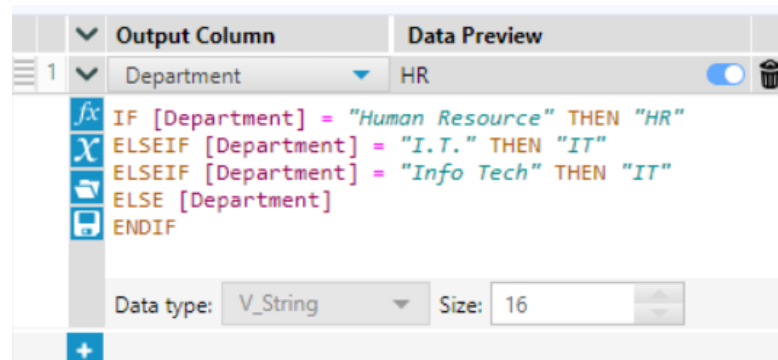
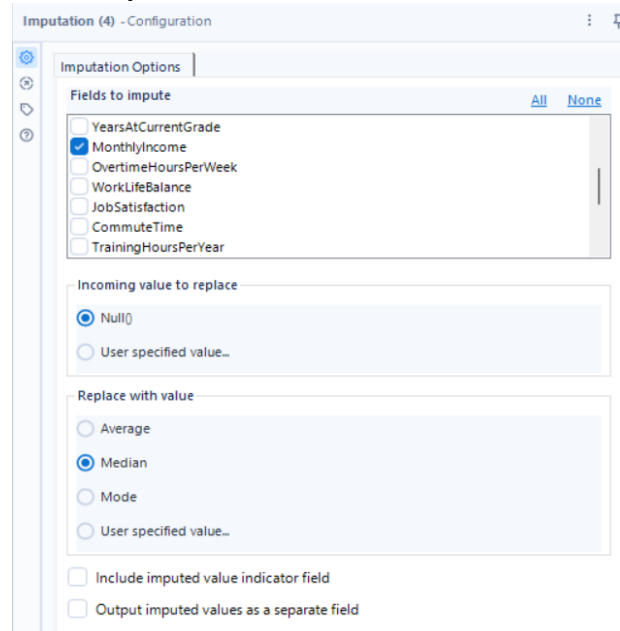
Distinct Values

Finance	853
Operations	838
Customer Service	802
Sales	798
HR	588
I.T.	433
IT	344
Human Resource	255
Info Tech	89

After loading the data using the input tool, I used auto field to convert the data types into appropriate ones before further analysis. The tools field summary and browse were mainly used to explore and discover any data quality issues. The first issue I found was that the column MonthlyIncome had 0.24 percent of missing values. All the other columns had no

null values. The next issue I encountered was the inconsistencies in the Department column. The values HR and Human Resource referred to the same group. Likewise, IT, I.T and Info Tech are essentially all the same thing, just in different wordings and representation.

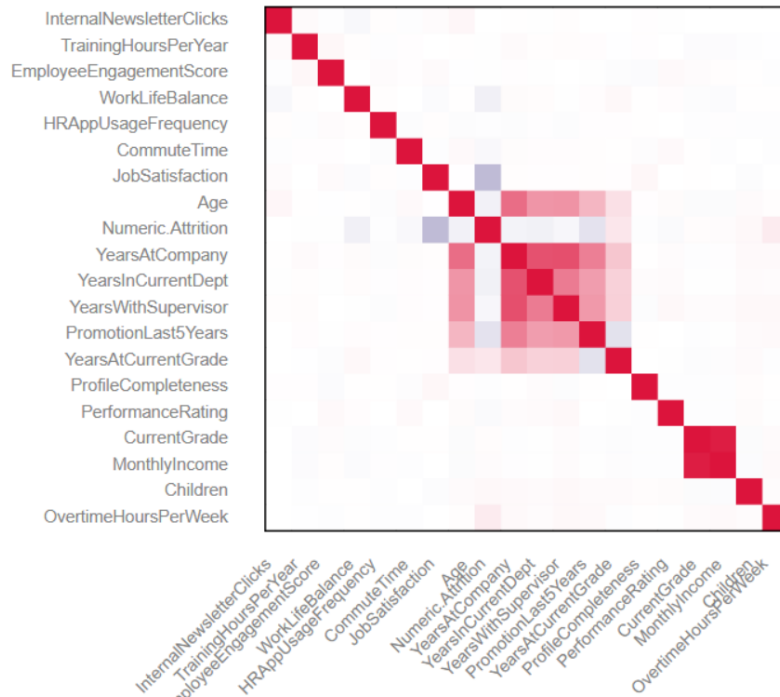
Data Preparation



My workflow starts with the input tool to get the data, followed by auto field to ensure appropriate data types. The imputation tool was then used to handle the missing values in the MonthlyIncome column. I imputed the null values with the median as the missing percent of 0.24 is very low and median is robust to outliers. This prevents losing the data and avoids any bias from using overall mean. Next, to fix the inconsistency issue in the Department column, I used the formula tool to group up the HR and IT departments instead of having them separated as shown in the picture above. This ensures consistent grouping for further analysis, modelling and reporting.

3. Feature Preparation

Feature Selection Techniques:



Results - Pearson Correlation (14) - Output

21 of 21 Fields | Cell Viewer | 20 records displayed | Search | Data | Metadata | Actions | 000

Record	FieldName	Age	Children	YearsAtCompany	YearsInCurrentDept	YearsWithSupervisor	CurrentGrade	Ye
11	JobSatisfaction	-0.002517	-0.015972	0.000588	0.007202	0.002481	0.006747	0.0
12	CommuteTime	0.023937	0.008103	0.015607	0.014109	0.02128	0.005048	0.0
13	TrainingHoursPerYear	0.006865	-0.015536	0.019062	0.007139	0.016557	-0.019141	-0.0
14	PerformanceRating	0.018989	0.019701	0.016811	0.023948	0.030262	-0.009377	-0.0
15	PromotionLast5Years	0.30592	0.023934	0.434223	0.31883	0.326656	-0.005623	-0.0
16	EmployeeEngagementScore	-0.002688	0.003626	0.014833	0.005571	0.013562	0.014981	-0.0
17	InternalNewsletterClicks	0.036697	-0.003317	0.007374	-0.010373	0.001832	-7.9e-05	-0.0
18	ProfileCompleteness	0.007397	0.01086	0.004045	0.018119	-0.013936	-0.011791	0.0
19	HRAppUsageFrequency	-0.017231	-0.010576	-0.01032	0.00924	-0.012481	-0.006946	-0.0
20	NumericAttrition	-0.070693	0.024791	-0.074813	-0.076208	-0.057573	0.013416	0.1

Record Layout

1

Table of Gender by Attrition

Gender	Type	Attrition: No	Attrition: Yes	Total
Female	Frequency	1895	648	2543
	Percent	37.9	12.96	50.86
	Row Percent	74.52	25.48	-
	Column Percent	49.49	55.34	-
Male	Frequency	1934	523	2457
	Percent	38.68	10.46	49.14
	Row Percent	78.71	21.29	-
	Column Percent	50.51	44.66	-
Total	Percent	76.58	23.42	100
	Frequency	3829	1171	5000

Chi-squared = 12.2650, df = 1, p-value < 0.0005

Table of EducationLevel by Attrition

EducationLevel	Type	Attrition: No	Attrition: Yes	Total
Diploma	Frequency	1138	359	1497
	Percent	22.76	7.18	29.94
	Row Percent	76.02	23.98	-
	Column Percent	29.72	30.66	-
Master	Frequency	473	128	601
	Percent	9.46	2.56	12.02
	Row Percent	78.7	21.3	-
	Column Percent	12.35	10.93	-
NITEC	Frequency	191	49	240
	Percent	3.82	0.98	4.8
	Row Percent	79.58	20.42	-
	Column Percent	4.99	4.18	-
Degree	Frequency	1897	594	2491
	Percent	37.94	11.88	49.82
	Row Percent	76.15	23.85	-
	Column Percent	49.54	50.73	-
PhD	Frequency	130	41	171
	Percent	2.6	0.82	3.42
	Row Percent	76.02	23.98	-
	Column Percent	3.4	3.5	-
Total	Percent	76.58	23.42	100
	Frequency	3829	1171	5000

Chi-squared = 3.2605, df = 4, p-value < 0.5152

The method I used for feature selection is using the Association analysis and Pearson correlation for numeric variables to measure their linear relationship with the target variable, attrition. As attrition in the dataset is a categorical variable with yes/no, I first had to encode into a 0/1 where yes = 1 and 0 = no. A new column, numeric attrition was derived. This was used as the target variable for Pearson correlation. From my findings, the highest correlated variable is JobSatisfaction with $r = -0.352$. It is the strongest negative and low satisfaction is strongly linked to attrition. YearsAtCurrentGrade has a $r = 0.177$ which is positive, and prolonged years can increase the risk of leaving. PromotionLast5Years has a $r = -0.131$ which is negative, and lack of promotion can increase the risk of leaving.

Features with near zero correlations were considered for exclusion as they seem to have very minimal or no impact on attrition. Anything below the 0.03 mark was the decider in excluding the variables as they are the least important.

For categorical columns, I used contingency table and apply chi-squared statistic to check which features are relevant. A p-value of less than 0.5 means that that feature is significant. Domain knowledge was also used for categorical features.

Features selected:

Age was included as it had a normal negative correlation with -0.071. This indicates that younger employees tend to leave more often. Gender, MaritalStatus and Children are kept as certain values in each column can have effects on attrition. For example, single employees may be more likely to leave etc. Department and Job role are also important as they give valuable information on what role is the person under and we can see which department has the most attrition. Columns such as YearsAtCompany, YearsInCurrentDept,

YearsWithSupervisor, CurrentGrade, YearsAtCurrentGrade are kept as they all had suitable correlation value. MonthlyIncome and OvertimeHoursPerWeek also show good correlation value. WorkLifeBalance and JobSatisfaction are on the higher side of correlation and low scores can indicate signs that someone may leave. PromotionLast5Years and MentorshipProgramOptIn indicates no promotion or not joining the mentorship increases risk.

Features removed:

EmployeeID was removed as it is only used as an identifier and has no predictive value for modelling or any relationship to attrition at all. EducationLevel was removed as the chi-square result showed p value more than 0.5 and domain research shows that education level is not a primary indicator for attrition. TrainingHoursPerYear had a very low pearson correlation, -0.006 and had no significant relation with attrition and was removed. Similarly, EmployeeEngagementScore also had the same correlation and was removed.

PerformanceRating also had a low correlation of -0.025 and was removed.

InternalNewsletterClicks, ProfileCompleteness, HRAppUsageFrequency and Commutetime all had very weak correlation near 0 and were removed.

Research on Machine Learning Algorithms

To begin with, Logistic Regression is a commonly used classification model that looks at the relationship between independent variables and log-odds of a binary outcome. It normally assumes a linear relationship between the predictors and logit of the target variable. To prepare for this model, categorical variables need to be one-hot encoded to convert them into numeric dummy variables.

For Decision Tree, they have no assumption on linear and can handle mixed data types, be it numeric or categorical. This makes them useful for pointing out complex and non-linear relationships between features and attrition. However, they are prone to overfitting if not pruned, especially if the model is allowed to grow too deep. Not much is needed to prepare for this model, only pruning parameters of minimum leaf size to prevent overfitting the dataset and improve performance.

Neural network is another machine learning algorithm to model complex and non-linear relationships within employee characteristics data. Similar to logistic regression, this model requires all features to be numeric, which needs one-hot encoding to be implemented before using the model.

Domain Research and Feature Engineering

From the domain research of reliable HR sources, they have identified some common predictors of attrition. These include low job satisfaction, excessive overtime, no promotions, and younger employees who seek other opportunities to grow. This information helped me to prepare for my feature engineering. In total, I created 3 new features using the formula tool. Firstly, I created a feature called Workload Stress which finds out the effect of high overtime when the work-life balance is poor, which reflects burnout by the individual and it may not be easy to tell from the other variables alone. Satisfaction Composite is another feature that takes the average of Job Satisfaction and

Work-life balance for a stronger and more robust predictor and reduces multicollinearity. Lastly, I created a Promotion Flag feature that tells us about any recent promotion and no promotion for models to better interpret and give better results. Normally, if the employee gets one promotion over the last 5 years, it should keep them stable at the job as they have good progression. That's why this feature was added to check whether they have had any promotion over the last few years and use that for the models.

The image displays three sequential screenshots of the 'Formula (15) - Configuration' window, showing the configuration of three new output columns.

First Screenshot: WorkloadStress

- Output Column:** WorkloadStress
- Data Preview:** 0
- Formula:** `[OvertimeHoursPerWeek] * (5 - [WorkLifeBalance])`
- Data type:** Byte
- Size:** 1

Second Screenshot: PromotionFlag

- Output Column:** PromotionFlag
- Data Preview:** 0
- Formula:** `IF [PromotionLast5Years] > 0 THEN 1 ELSE 0 ENDIF`
- Data type:** Byte
- Size:** 1

Third Screenshot: SatisfactionComposite

- Output Column:** SatisfactionComposite
- Data Preview:** 5
- Formula:** `([JobSatisfaction] + [WorkLifeBalance]) / 2`
- Data type:** Int16
- Size:** 2

Consideration of Unsupervised Machine Learning Models

In addition to supervised learning methods, unsupervised machine learning is essential in revealing any relationships that may not be so clear and discovering any hidden patterns. Clustering is a model that is valuable for a HR context and can help to segment employees into distinct groups based on key features like age, promotion flag etc. This will allow target intervention strategies for different risk profiles. This method is suitable for the dataset with its mixture of numeric and categorical features. For algorithm, K-means was selected as it is simple yet efficient on a 5000-row dataset and it is also able to handle numeric data. Euclidean distance is suitable for numeric features. The number of clusters can be explored between 4-5 which is ideal as selecting too few clusters can oversimplify the meaningful differences between the clusters, and adding too many clusters can result in noisy and less interpretable clusters. This range provides a good balance for pattern discovery which allows to identify distinct employee segments with different attrition risks.

4. Model Building

Partitioning the Data:

Simulation Sampling (43) - Configuration

Simulation Configuration

Select sampling mechanism

Monte Carlo/ Simple Sampling

Chunk size

256000

Seed

1

Number of iterations

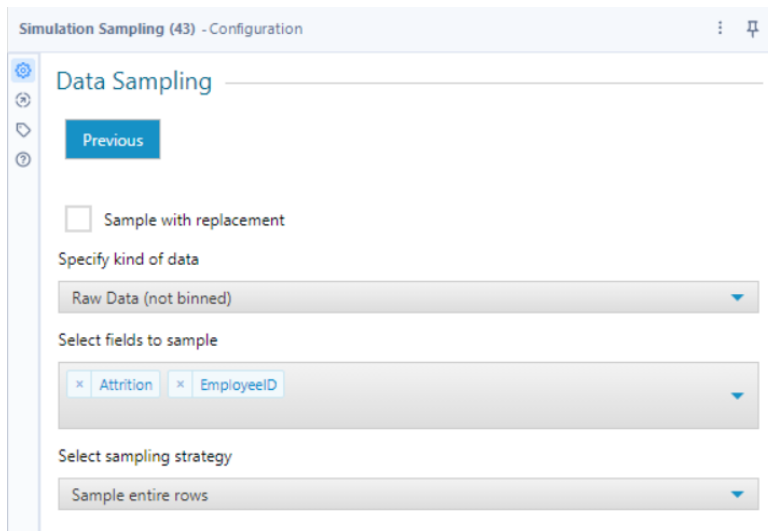
3500

Select sampling mode

☐ Sample parametrically

☒ Sample from data

Next



After one hot encoding of categorical variables, the data was partitioned using the Simulation Sampling tool where I trained 70% of the data (3500 rows) and 30% of the data for testing (1500 rows). This partitioning ensures fairness and not random sampling as there is around 23% of employees with “Yes” for attrition. Random sampling and splitting may result in the train to have bias or can over/under fit the model. Stratified Sampling ensures a good generalisation and unbiasedness for machine learning.

Modelling Techniques

Two machine learning models were used for this context. Logistic regression and Decision tree. These models were chosen for balanced predictive performance.

The first model I used was Logistic Regression. It can handle binary outcomes which is much suited for the target variable, attrition, and handles numeric and encoded categorical variables. This model tells us the coefficients of each factor and clearly identifies its impact on the target variable. This will allow the HR to easily understand why an employee may be at higher risk of leaving. This allows them to make data-driven decisions rather than relying on complex predictions.

The second model I used was Decision Tree. It can handle mixed data types and non-linear relationships. It produces simple rules such as if a feature shows low output and has no recent promotion, it will lead to high risk. This visual will help more in terms of prediction and allows us to get further insights on what each feature does in relation to attrition. Both models give predictive accuracy and allows HR to understand why the employees are at risk and not just who.

By using both models, analysis can be done with predictive accuracy with business interpretability, ensuring that HR can understand the reasoning behind each employee’s attrition.

Training and Tuning of Models

Logistic Regression

Models were trained using the 70% training set and tuned to improve performance. For my first logistic regression model, I set the target variable as attrition and included all the prepared features, both numeric and one-hot encoded. From there, I found a few variables which had no impact at all and some variables that were also not very useful. The variables in the screenshot below show which had no impact. After looking at p-values around 0.6 and above, I removed those for better model performance. For the second, final and tuned logistic model, it contains 24 variables, including some of the one-hot encoded. See attached screenshot for the selected variables.

Removed features:

MaritalStatus_Widowed	NA	NA	NA	NA
Department_Sales	NA	NA	NA	NA
JobRole_Finance.Manager	NA	NA	NA	NA
JobRole_Operations.Manager	NA	NA	NA	NA
JobRole_Team.Lead	NA	NA	NA	NA
JobRole_infrequent_sklern	NA	NA	NA	NA

Children (0.967), YearsAtCompany (0.535), CurrentGrade (0.950), MentorshipProgramOptInYes (0.986), DepartmentHR (0.913), JobRole_Customer.Support (0.872), JobRole_HR.Executive (0.916), JobRole_Operations.Analyst (0.720)

Final selected features:

☒ Select All

☐ EmployeeID
☒ Age
☐ Children
☐ YearsAtCompany
☒ YearsInCurrentDept
☒ YearsWithSupervisor
☐ CurrentGrade
☒ YearsAtCurrentGrade
☒ MonthlyIncome
☒ OvertimeHoursPerWeek
☒ WorkLifeBalance
☒ JobSatisfaction
☒ PromotionLast5Years
☐ MentorshipProgramOptIn
☐ Numeric Attrition
☒ WorkloadStress
☒ PromotionFlag
☒ SatisfactionComposite
☒ Gender_Male
☒ MaritalStatus_Divorced
☒ MaritalStatus_Married
☒ MaritalStatus_Separated
☒ MaritalStatus_Single
☐ MaritalStatus_Widowed
☒ Department_Customer Service
☒ Department_Finance
☐ Department_HR
☒ Department_IT
☒ Department_Operations

☐ Department_Sales
☒ JobRole_Accountant
☐ JobRole_Customer Support
☐ JobRole_Finance Manager
☐ JobRole_HR Executive
☐ JobRole_Operations Analyst
☐ JobRole_Operations Manager
☒ JobRole_Sales Executive
☒ JobRole_Software Developer
☐ JobRole_Team Lead
☐ JobRole_infrequent_sklern

Decision Tree

For my first model, I set the target variables is attrition and used rpart as the algorithm. Model type is classification and split using Gini impurity. When pruning, I removed the complexity parameter.

Splitting Criteria and Surrogates

The splitting criteria to use:

☒ Gini impurity

☐ Information gain

Use surrogates to

☒ Omit observations with missing value for primary split rule

☐ Split records missing the candidate variable

☐ Send observation in majority direction if all surrogates are missing

Select best surrogate split using

☒ Number of correct classifications for a candidate variable

☐ Percentage of correct classifications for a candidate variable

HyperParameters

The minimum number of records needed to allow for a split

20

The allowed minimum number of records in a terminal node

7

The number of folds to use in the cross validation to prune the tree

10

The maximum allowed depth of any node in the final tree

20

The maximum number of bins to use for each numeric variable

Default

☒ Set complexity parameter 0

Important Variables

Logistic regression provided the coefficients and odds ratio to help us determine the important variables that can help to predict the target variable. Below are the top 5 variables that have the top few coefficients. The results show that an employee's well-being, career progression and how satisfied they are at work have key roles leading to attrition.

Variable	Coefficient	P-value
JobSatisfaction	-0.8621	< 2.2e-16
YearsAtCurrentGrade	0.2760	< 2.2e-16
PromotionFlag	-1.151	2.55e-13
WorkloadStress	0.04085	4.63e-10
Gender_Male	-0.3357	0.00027

Exploration of Different Approaches

Clustering was used as an Unsupervised machine learning method to uncover hidden employee segments that may not be easily seen through attrition prediction alone. Clustering can help to further explain in detail why certain groups of employees with similar profiles have risk of leaving. This aligns with the business objective of providing HR with deeper insights to support targeted retention strategies.

Firstly, I used the K-Centroids Cluster Analysis tool as it is suited for employee segmentation problems and is efficient for this dataset. Clustering does not need any train-test split, so it was done on the full dataset so that it can benefit by learning patterns across 100% of records.

Feature Selection and Data Preparation for Clustering

K-Centroids Cluster Analysis (37) - Configuration

Configuration | Plot Options | Graphics Options

Solution name

Fields (select two or more) [All](#) [None](#)

- ☒ Age
- ☒ Children
- ☒ YearsAtCompany
- ☒ YearsInCurrentDept
- ☒ YearsWithSupervisor
- ☒ CurrentGrade
- ☒ YearsAtCurrentGrade

☒ Standardize the fields

☐ z-score

☒ Unit interval

Clustering method

☒ K-Means

☐ K-Medians

☐ Neural Gas

Number of clusters

4

Number of starting seeds

10

A total of seven features was selected based on their domain relevance and their association with factors like employee well being and workload. These included Workload Stress, Satisfaction Composite, Overtime Hours Per Week, Job Satisfaction, Years at Current Grade, Promotion Flag and Age. These variables can help to capture and segment them by values like stress, workload intensity etc.

Selection of Number of Clusters

The number of clusters was set to K = 4, which provides a balance between interpretability and meaningful segmentation. A fewer k value can mask important differences, while a larger k value can result in too much and wide of a group which can be hard to interpret and make actions. This value gave us stable clusters after 38 iterations, which is reliable.

Cluster Performance

Record	Report																									
1	Summary Report of the K-Means Clustering Solution X																									
2	<i>Solution Summary</i>																									
3	Call: stepFlexclust(unitScale(model.matrix(~-1 + Age + Children + YearsAtCompany + YearsInCurrentDept + YearsWithSupervisor + CurrentGrade + YearsAtCurrentGrade + MonthlyIncome + OvertimeHoursPerWeek + WorkLifeBalance + JobSatisfaction + PromotionLast5Years + WorkloadStress + PromotionFlag + SatisfactionComposite, the.data)), k = 4, nrep = 10, FUN = kcca, family = kccaFamily("kmeans"))																									
4	Cluster Information:																									
5	<table><tr><th>Cluster</th><th>Size</th><th>Ave Distance</th><th>Max Distance</th><th>Separation</th></tr><tr><td>1</td><td>1141</td><td>0.810965</td><td>1.473855</td><td>0.544204</td></tr><tr><td>2</td><td>856</td><td>0.886359</td><td>1.561124</td><td>0.621962</td></tr><tr><td>3</td><td>1800</td><td>0.922557</td><td>1.850129</td><td>1.165754</td></tr><tr><td>4</td><td>1203</td><td>0.819638</td><td>1.57094</td><td>0.622049</td></tr></table>	Cluster	Size	Ave Distance	Max Distance	Separation	1	1141	0.810965	1.473855	0.544204	2	856	0.886359	1.561124	0.621962	3	1800	0.922557	1.850129	1.165754	4	1203	0.819638	1.57094	0.622049
Cluster	Size	Ave Distance	Max Distance	Separation																						
1	1141	0.810965	1.473855	0.544204																						
2	856	0.886359	1.561124	0.621962																						
3	1800	0.922557	1.850129	1.165754																						
4	1203	0.819638	1.57094	0.622049																						
6	Convergence after 38 iterations. Sum of within cluster distances: 4330.66154.																									

It resulted in a total of 4 clusters with sizes ranging from 856 to 1800 employees. The average distance was relatively consistent across the 4 clusters suggesting cohesion within each group. Cluster 3 had the highest separation value of 1.165, suggesting that it was the most distinct group in the dataset.

Interpretation of Clusters

After using K-Centroids Cluster Analysis and Append Cluster, I used crosstab tool to group the data by cluster with column header being attrition and aggregated the values by using count and percent row. This gave me an output of the count and percent of how many “No” and “Yes” in each cluster.

Record	Cluster	XRow_No	Count_No	XRow_Yes	Count_Yes
1	4	88.560335	1060	11.439665	143
2	2	80.628356	684	19.371644	172
3	1	78.840764	894	21.159236	247
4	3	65.769473	1191	34.230527	609

Cluster 3 showed the highest attrition rate with 34.2% of employees leaving. This cluster contains characteristics like not getting promotion or lesser promotion exposure, higher workload stress and lower job satisfaction levels, suggesting a burnout- employees segment. Cluster 1 and 2 showed moderate attrition rates of around 21.2% and 19.4% respectively. The employees in these clusters also have moderate and mixed satisfaction levels and limited career progression, which can cause them to be a potential risk of attrition if not managed. Lastly, Cluster 4 showed the lowest attrition rate with 11.4%. This cluster contains employees with higher satisfaction scores, better work-life balance and lower workload stress. This is the best cluster which shows that they are the most engaged group.

Cross Tab (60) - Configuration

Filter by Keyword

- ☐ JobRole_HR.Executive
- ☐ JobRole_Operations.Analyst
- ☐ JobRole_Operations.Manager
- ☐ JobRole_Sales.Executive
- ☐ JobRole_Software.Developer
- ☐ JobRole_Team.Lead
- ☐ JobRole_infrequent_sklearn
- ☒ Cluster

Change Column Headers

Attrition

Values for New Columns

Age

☐ Retain Special Characters in New Column Names

Method for Aggregating Values

- ☐ Average
- ☒ Count (Without Nulls)
- ☐ Count (With Nulls)
- ☐ Count Distinct (Without Nulls)
- ☐ Count Distinct (With Nulls)
- ☒ Percent Row
- ☐ Percent Column

5. Model Evaluation

Metrics and Evaluation

For evaluation of models, accuracy which shows overall correct predictions, accuracy of yes, precision of yes which shows the proportion of predicted yes that are actual yes, recall of yes which shows the proportion of actual correctly predicted yes, F1-score and AUC (area under the ROC curve)

Model	Accuracy	Accuracy_Yes	Precision	Recall	F1	AUC
Logistic Regression	0.7873	0.3128	0.605	0.313	0.4125	0.8163
Decision Tree	0.8100	0.5670	0.610	0.567	0.5876	0.8116

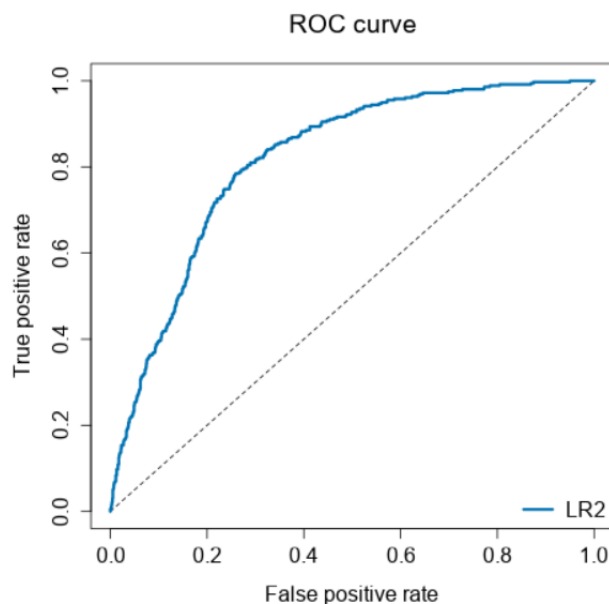
Logistic Regression Metrics

Model	Accuracy	Accuracy_No	Accuracy_Yes	F1	AUC
LR2	0.7873	0.9361	0.3128	0.4125	0.8163

Confusion matrix of LR2

	Actual_No	Actual_Yes
Predicted_No	1069	246
Predicted_Yes	73	112

ROC Curve plots the true positive rate against the false positive rate at all possible decision thresholds. The ROC curve for the tune Logistic Regression model shows a smooth upward curve starting from (0,0) bending towards the (1,1) with AUC being around 0.82 from the metric. This indicates a good model in being able to identify correctly for ranking a random leaver higher in risk than a random stayer. The curve has a sharp increase in the left side of the graph which means the model can achieve high true positive rate without many false positives. For example, at FPR = 0.2, TPR can be around 0.6 which means 60% of leavers are caught with 20% false alarms.

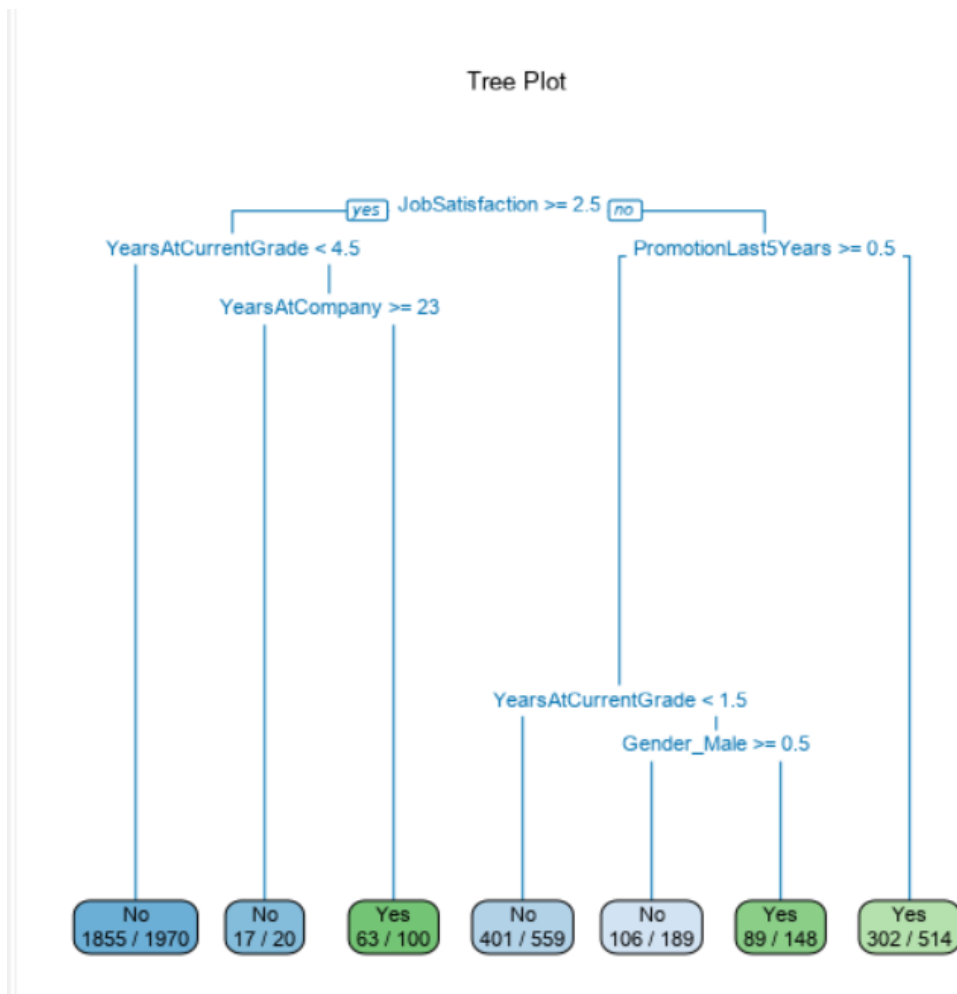


Decision Tree Metrics

Model	Accuracy	Accuracy_No	Accuracy_Yes	F1	AUC
DT2	0.8100	0.8862	0.5670	0.5876	0.8116

Confusion matrix of DT2

	Actual_No	Actual_Yes
Predicted_No	1012	155
Predicted_Yes	130	203



Above is the tree plot from the decision tree. The most obvious red flag is at the top where Job Satisfaction is lower than or equal to 2.5, it automatically moves employees to the high-risk side. Moving on, if they do not have a promotion over the last 5 years, less than 0.5, it worsens the risk. Having high job satisfaction almost guarantees that the employee might stay as very little of them have left. The tree using only three main and top features ensures simplicity and makes it very practical for HR.

Trade-offs between Metrics

The first trade-off is between accuracy and recall for “Yes”. In this dataset, majority of the employees are labelled with “No” for attrition. This means that a high accuracy is still possible even if the model does not undergo any learning and if it were to simply predict no for everyone. For this problem statement, accuracy is not the best metric to look at. Instead, recall is better as it can be used to identify employees who might resign early. A high accuracy model with low recall can miss most leaver and makes it inefficient. Therefore, prioritising accuracy can risk missing employees that leave while prioritising recall might reduce overall accuracy possibly due to more false negatives. The second trade-off is between precision and recall for “Yes”. Precision identifies how many of the predicted employees to leave are leaving. High precision means that there are very few errors in predicting. However, high precision can mean low recall. The model will only flag the most obvious cases and may leave out other actual leavers. Similarly, high recall will increase the false positives and catch more of the actual leavers. In this case, recall should be prioritised.

as the consequence of missing someone that leaves is worse than the consequence of a false alarm.

Research on Advanced Model Evaluation Techniques

One advanced model evaluation technique is the ROC Curve and AUC. The ROC curve (receiver operating characteristics) will plot the true positive rate against the false positive rate across all possible decision thresholds. AUC will summarize the value between 0-1 and measure the model's ability. I chose AUC as the primary measure as it evaluates the ranking quality which is essential for HR to prioritise high risk employees. Lift chart measures how good the model is in identifying "Yes" cases in the top few percentages. For example, a lift of 3 predicts that the top 10% of predicted risk contain 3 times more employees who leave than expected. Lift can be useful for business applications like targeted retention

6. Key Insights and Recommendation

Key Insights:

The number one predictor is Job Satisfaction. Logistic Regression results showed that having a low score can increase the odds of the employee leaving. Even if other factors are normal, being dissatisfied with the job can be a key reason as to why they want to leave. This confirms that even if employees are productive and engaged, they can still resign.

Additionally, the year at current grade also shows a strong positive relationship. Employees that remain in the same grade for longer periods or have not received any promotion, are likely at higher risk of leaving. This suggests that lack of career progression plays a vital role when it comes to attrition. This finding supports HR that link career development opportunities to employee retention.

The engineered feature, workload stress which takes overtime hours and poor work-life balance, is also highly significant. High overtime and low work life balance can cause burnout which strongly predicts attrition. This explains why some employees with moderate individual score can leave unexpectedly as the cumulative effect of workload-related stress is captured more effectively through this composite feature.

Clustering revealed that the third cluster is identified as the highest risk. This cluster is characterised by lower satisfaction levels, low promotion exposure, higher stress and workload. These are the key factors to look in when identifying employees with high risk.

Recommendations:

First thing that can be done is to monitor and survey the employees regularly about their Job satisfaction. Employees that score 2 and below should be flagged for deeper investigation such as one-on-one meetings to understand them and look for areas to improve. Early intervention allows HR to identify any issues before the employee is dissatisfied and goes to the extent of leaving.

Next, they need to prioritize promotion and progression for employees. Those that have no promotion for 5 years or are in the same grade for more than 3-4 years, need to be prioritized for any developments or role changes. HR can prevent this risk by ensuring employees get development opportunities such as upskilling programs to maintain employee engagement.

Third, overtime needs to be adjusted for employees so that they can have a better work life balance and not be too stressed about the workload. New arrangements need to be made to make it manageable for them and prevent them from doing too much overtime. Workloads need to be redistributed; flexible working arrangements need to be made or can improve staffing allocation. Reducing overtime can help prevent burnout and improve long term retention.

Lastly, HR can make use of the clustering results to design any segment specific intervention strategies. They need to target the third cluster which has the highest risk and implement any intensive engagement or workload interventions. This cluster should be prioritized, and lower risk cluster may require minimal monitoring.

Limitations and Improvements:

We need to take note that the analysis is based on historical data and cannot adhere to any other factors such as their relationship with manager, work culture or personal circumstances. The dataset only represents data at one point of time and may not capture any changes in employee characteristics over longer periods. For future improvements, real-time monitoring can be done to enhance predictive accuracy. Keep the machine learning models updated with new data to ensure it can continuously work in predicting the reasons and those who are at risk of attrition.

[Tawfeeq]



References

[Include any websites, videos, books, articles, reports you referenced]